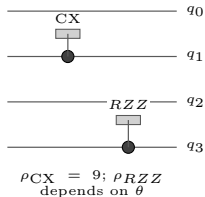


CERTICUT: Representation-Sensitive Placement with Resource Bounds

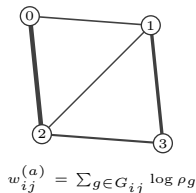
Fixed logical circuit and feasible placement space $\mathcal{F}$

Fixed K , capacities (L_k, U_k) , logical wires, and admissible fragment maps

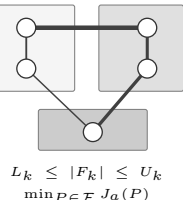
(a) Circuit a



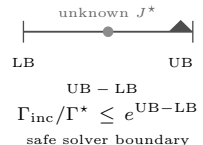
(b) Graph $G^{(a)}$



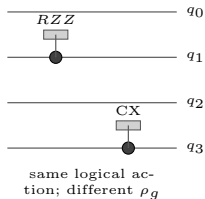
(c) K -way partition



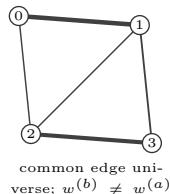
(d) Resource bound



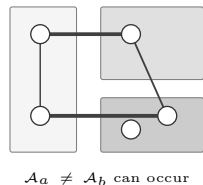
(e) Circuit b



(f) Graph $G^{(b)}$



(g) New partition



(h) Sensitivity

$$\begin{aligned} \mathcal{A}_a &= \arg \min_P J_a(P) \\ \mathcal{A}_b &= \arg \min_P J_b(P) \end{aligned}$$

Reversal / regret

$$\Delta_{a \rightarrow b} > 0 \Rightarrow R_{a \rightarrow b} > 1$$

Stability

$$m_a > \|\delta\|_1$$

$$\mathcal{A}_b \subseteq \mathcal{A}_a$$

Edge width denotes $\log \rho$; node set and feasible placement space remain fixed across representations.